

Solution of exercises lecture 10

1 Show that pricing this option with the risk-neutral valuation principle give different results than with real-world probabilities.

We have the following:

$$S_0 = 100, \quad S_u = 150, \quad S_d = 75, \quad p = \frac{1}{2}, \quad r = 0, \quad K = 110.$$

The payoff of the call at maturity is

$$C_u = \max(150 - 110, 0) = 40 \quad \text{and} \quad C_d = \max(75 - 110, 0) = 0.$$

The price obtained using the real-world probability $p = \frac{1}{2}$ is

$$C_0^{\text{real}} = \frac{1}{2}(40) + \frac{1}{2}(0) = 20.$$

We search for the risk-neutral probability q using

$$S_0 = qS_u + (1 - q)S_d.$$

Plugging in the values gives

$$\begin{aligned} 100 &= 150q + 75(1 - q) \\ \iff 25 &= 75q \\ \iff q &= \frac{1}{3} \end{aligned}$$

The risk-neutral price of the call is therefore

$$C_0^{RN} = \frac{1}{3}(40) + \frac{2}{3}(0) = \frac{40}{3} = 13.33$$

This differs from the value obtained using real-world probabilities.

2 Show how you could construct an arbitrage strategy if someone would sell you the call at the price computed with real-world probabilities.

We replicate the option using:

- Δ shares of stock,
- α units invested in the risk-free asset.

The portfolio must satisfy

$$150\Delta + \alpha = 40 \quad \text{and} \quad 75\Delta + \alpha = 0.$$

Subtracting the second equation from the first:

$$75\Delta = 40.$$

$$\Longleftrightarrow \Delta = \frac{40}{75} = \frac{8}{15} \approx 0.5333.$$

Substituting back:

$$\alpha = -75\Delta = -40$$

Thus the initial cost is

$$0.5333 \cdot 100 - 40$$

$$= 53.33 - 40 = 13.33.$$

which matches the risk-neutral price.

Suppose someone sells the call option for the price computed using real-world probabilities $C_0 = 20$. However, the replicating portfolio only costs 13.33. Therefore, the option is overpriced by 6.67. So, at time 0, we could sell one call option for 20, and buy the replicating portfolio for 13.33. This generates an immediate risk-free profit of 6.67.